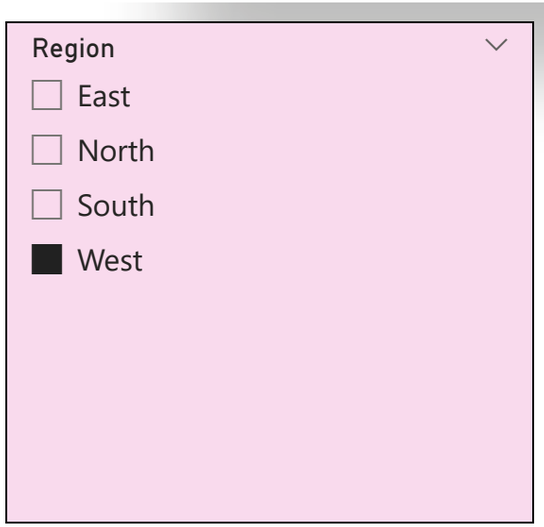
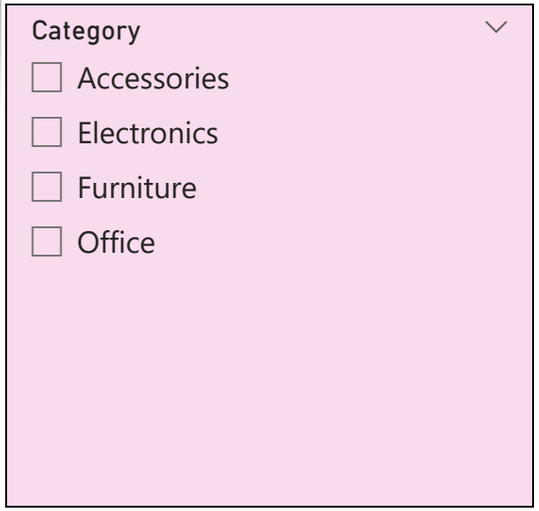
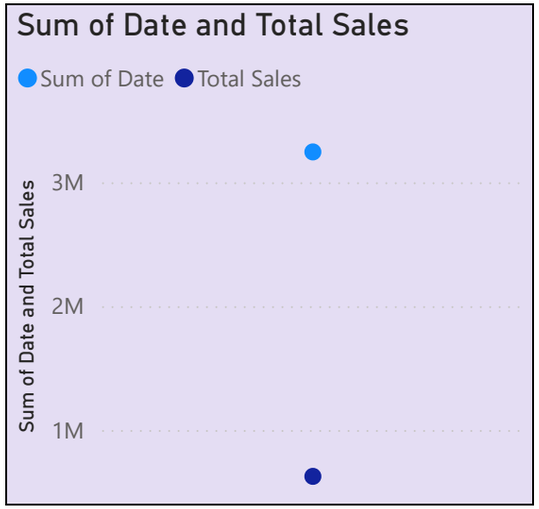
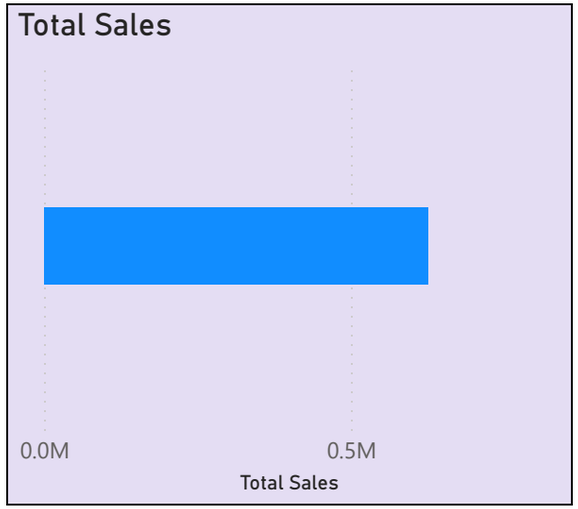
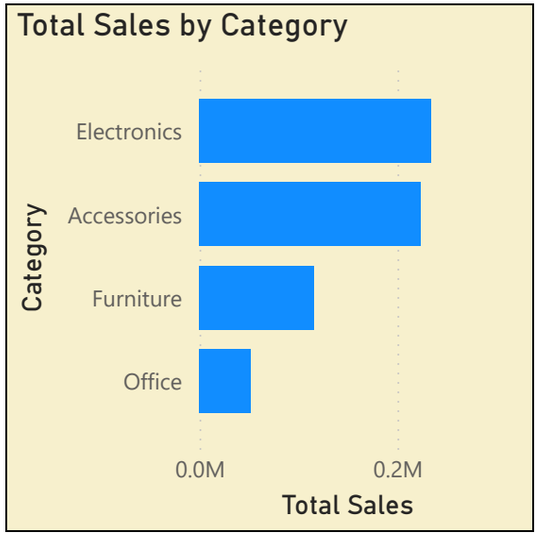
Assignment: Advanced Visuals in Power BI

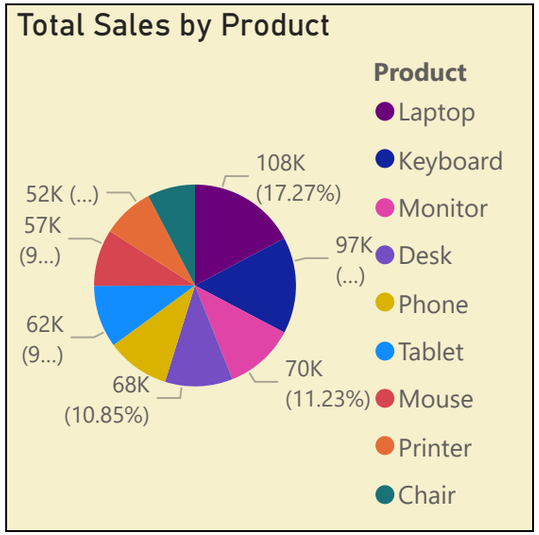




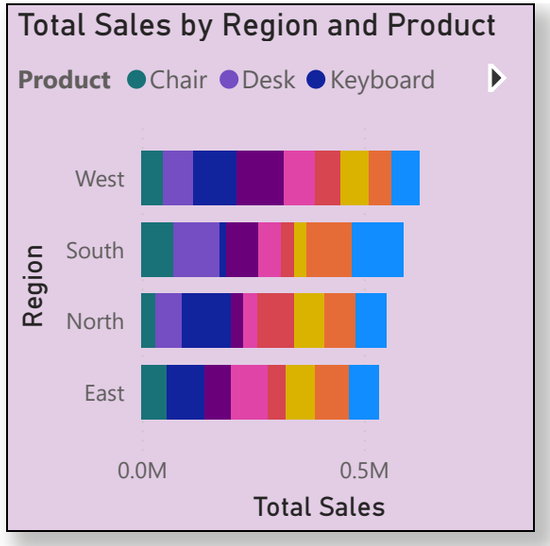








Product
Chair
Desk
Keyboard
Laptop
Monitor
Mouse
Phone
Printer
Tablet



1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Basic Filter
A basic filter allows users to filter data by selecting values from a list. It is simple and used for direct selection of data values.

Advanced Filter
An advanced filter allows users to apply conditions such as greater than, less than, contains, or begins with. It is used for more complex filtering conditions.

Slicer
A slicer is a visual filtering tool placed on the report canvas that allows users to interactively filter multiple visuals in a report.

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2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

To apply a basic filter in Power BI:

1. Load the sales dataset into Power BI.
2. Select a visual or table.
3. Open the Filters pane.
4. Drag the **Year field** into the filter section.
5. Select the value **2023**.

This will display sales data only for the year 2023.

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3. Create a Pie Chart of Sales by Product and add a Slicer that allows users to filter the dataset by Region.

Steps:

1. Insert a **Pie Chart** visual in Power BI.
2. Drag the **Product field** to the Legend.
3. Drag the **Sales field** to Values.

4. Insert a **Slicer visual**.
5. Drag the **Region field** to the slicer.

Now users can filter the pie chart using the Region slicer.

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4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

A **Visual-Level Filter** affects only a single visual in the report.

A **Page-Level Filter** affects all visuals present on a specific report page.

A **Report-Level Filter** affects all visuals across the entire Power BI report.

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5. Apply a Report-Level Filter so that all visuals display sales only from India.

Steps:

1. Open the Filters pane.
2. Locate the **Report-Level Filters** section.
3. Drag the **Country field** into this section.
4. Select **India**.

Now all visuals in the report will show only sales data from India.

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6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

Drill-Down allows users to explore data at different levels of detail within the same visual. It enables users to move from summarized data to more detailed data.

For example, users can drill down from **Category to Product** to analyze sales performance at deeper levels.

This improves data exploration by allowing users to understand trends and patterns more clearly.

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7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales.

Steps:

1. In the Fields pane, right-click **Country** and select **Create Hierarchy**.
2. Add **Region** and **Product Category** to the hierarchy.
3. Use this hierarchy in a chart such as a bar chart.
4. Enable the **Drill-Down** feature to navigate through the levels.

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8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region-wise sales.

Steps:

1. Create a new report page.
2. Drag the **Product field** into the **Drill-Through filters section**.
3. Add a chart such as a bar chart.
4. Use **Region as Axis** and **Sales as Values**.

Now users can right-click a product and drill through to a detailed page showing region-wise sales.

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9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

Steps:

1. Insert a **Slicer visual**.
2. Drag the **Product Category field** into the slicer.
3. Select any category.

The drill-down charts will automatically update based on the selected category.

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10. Explain the difference between Drill-Down and Drill-Through using a real-world example.

Drill-Down allows users to move from higher-level data to lower-level data within the same visual.

Example: Sales → Category → Product.

Drill-Through allows users to navigate to another report page for detailed analysis.

Example: Clicking on **Laptop sales** opens a new page showing laptop sales by region.

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11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.

Steps:

1. Insert a **Line Chart**.
2. Drag the **Date field** to the Axis.
3. Power BI automatically creates a hierarchy: **Year → Quarter → Month → Day**.
4. Enable the **Drill-Down option** to analyze trends at different time levels.